

6 In a dividing cell, DNA replication occurs before mitosis.

(a) Steps in DNA replication are outlined in Fig. 6.1.

Complete Fig. 6.1 by filling in the gaps using the most appropriate terms.

1	Helicase enzyme allows the DNA double helix to unwind and the hydrogen bonds between the two strands to break, exposing the four bases, (A), (T), (C) and (G).
2	An enzyme molecule attaches to each of the two separated parental strands. The two enzyme molecules move in opposite directions, each catalysing the formation of a new strand of DNA. This enzyme is known as
3	DNA, the monomers of DNA, are activated with two additional phosphates and are free in the nucleus for the synthesis of the new strands.
4	The bases of the DNA monomers form hydrogen bonds with the bases on each separated parental strand of DNA, according to the rules of
5	One DNA strand is synthesised continuously and the other is synthesised in sections known as Okazaki fragments. The fragments are joined by an enzyme,, which catalyses the formation of phosphodiester bonds.
6	The result of replication is two DNA molecules, each one containing an original parental strand and a newly synthesised strand. This type of replication is described as

Fig. 6.1

(b) Fig. 6.2 is a photomicrograph of root tip cells at different stages in the cell cycle. A cell in interphase is labelled.

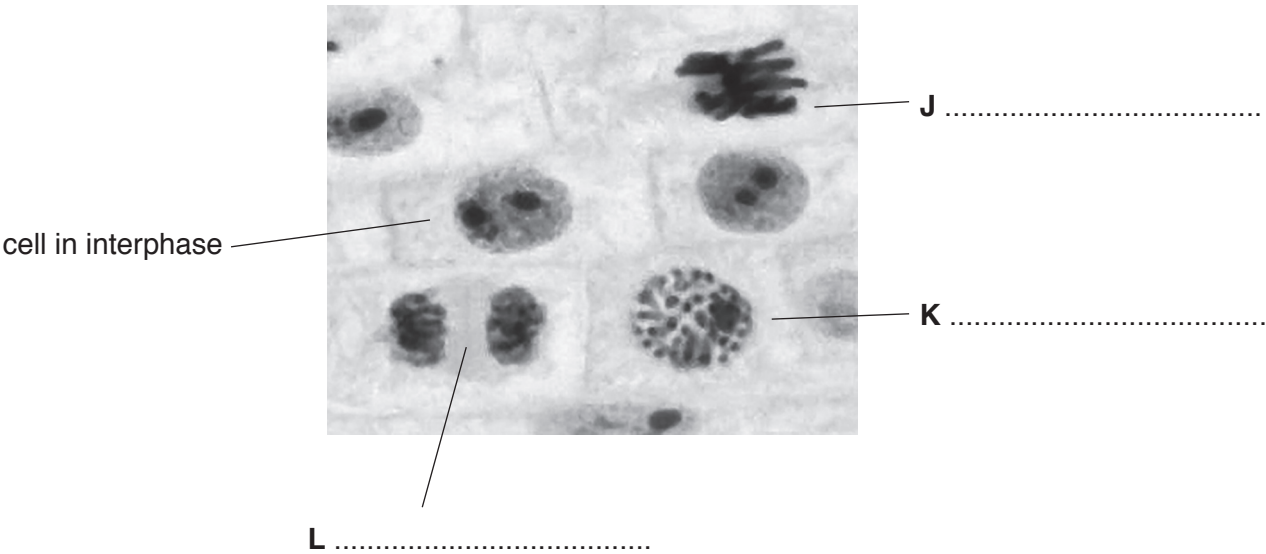


Fig. 6.2

(i) Name the stage of mitosis shown in each of cells **J**, **K** and **L** in Fig. 6.2.
 Write your answer in the space next to each letter on Fig. 6.2. [3]

(ii) Explain how it is possible to deduce that the labelled cell in interphase shown in Fig. 6.2 is in late, rather than early, interphase.

 [1]

(iii) Describe the stage of mitosis shown in cell **J**.

 [2]